



FA25-BCS-031

Inza bibi

Lab Assignment 1 — Part 1: ArrayList, Pointer Traversal & Problem Solving

```
#include <iostream>
```

```
using namespace std;
```

```
const int CAPACITY = 20;
```

```
struct ArrayList
```

```
{
```

```
    int data[CAPACITY];
```

```
    int size = 0;
```

```
};
```

```
ArrayList list;
```

```
int *ptr = list.data;
```

```
int minimum;
```

```
int maximum;
```

```
int median;
```

```
int closestValue;
```

```
int closestPosition;
```

```
int sum = 0;
```

```
double generalAverage;
```

```
double specialAverage;
```

```
double averageDifference;
```

```
double finalScore;
```

```
// ===== FUNCTIONS =====
```

```
bool insertEnd(ArrayList &list, int value) {
```

```
    if (list.size >= CAPACITY) {
```

```
        return false;
```

```
    }
```

```
    list.data[list.size] = value;
```

```
    list.size++;
```

```
    return true;
```

```
}
```

```
bool insertAtBeginning(ArrayList &list, int value) {
```

```
    if (list.size >= CAPACITY) {
```

```
        return false;
```

```
    }
```

```
    for (int i = list.size; i > 0; i--) {
```

```
        list.data[i] = list.data[i-1];
```

```
    }
```

```
    list.data[0] = value;
```

```
    list.size++;
```

```
    return true;
```

```
}
```

```

bool deleteAtPosition(ArrayList &list, int position) {
    if (position < 0 || position >= list.size) {
        return false;
    }
    for (int i = position; i < list.size - 1; i++) {
        list.data[i] = list.data[i+1];
    }
    list.size--;
    return true;
}

```

```

void displayList(const ArrayList &list) {
    for (int i = 0; i < list.size; i++) {
        cout << list.data[i] << " ";
    }
    cout << endl;
}

```

```

// ===== MAIN =====

```

```

int main() {

```

```

    // ----- PART A -----

```

```

    insertEnd(list, 18);

```

```

    insertEnd(list, 7);

```

```

    insertEnd(list, 45);

```

```

    insertEnd(list, 11);

```

```

    insertEnd(list, 36);

```

```
insertEnd(list, 31); //Reg No
```

```
insertEnd(list, 21);
```

```
insertEnd(list, 13);
```

```
insertEnd(list, 29);
```

```
cout << "Initial ArrayList: ";
```

```
displayList(list);
```

```
// ----- PART B -----
```

```
ptr = list.data;
```

```
sum = 0;
```

```
minimum = *ptr;
```

```
maximum = *ptr;
```

```
for (int i = 0; i < list.size; i++) {
```

```
    sum += *ptr;
```

```
    if (*ptr < minimum) minimum = *ptr;
```

```
    if (*ptr > maximum) maximum = *ptr;
```

```
    ptr++;
```

```
}
```

```
cout << "Minimum Value: " << minimum << endl;
```

```
cout << "Maximum Value: " << maximum << endl;
```

```
cout << "Sum: " << sum << endl;
```

```
// ----- PART C -----
```

```
int temp[CAPACITY];
```

```
for (int i = 0; i < list.size; i++) {
```

```
    temp[i] = list.data[i];
```

```
}
```

```
// Bubble Sort
```

```
for (int i = 0; i < list.size - 1; i++) {  
    for (int j = 0; j < list.size - i - 1; j++) {  
        if (temp[j] > temp[j+1]) {  
            int t = temp[j];  
            temp[j] = temp[j+1];  
            temp[j+1] = t;  
        }  
    }  
}  
}
```

```
median = temp[list.size / 2];
```

```
cout << "Median Value: " << median << endl;
```

```
// ----- PART D -----
```

```
generalAverage = (double)sum / list.size;
```

```
specialAverage = (double)(minimum + median + maximum) / 3;
```

```
ptr = list.data;
```

```
double minDistance = 999999;
```

```
closestPosition = 0;
```

```
for (int i = 0; i < list.size; i++) {  
    double distance = *ptr - specialAverage;  
    if (distance < 0) distance = -distance;  
  
    if (distance < minDistance) {
```

```
        minDistance = distance;

        closestValue = *ptr;

        closestPosition = i;
    }

    ptr++;
}
```

```
cout << "General Average: " << generalAverage << endl;
cout << "Special Average: " << specialAverage << endl;
cout << "Closest Value: " << closestValue << endl;
cout << "Position of Closest Value: " << closestPosition << endl;
```

```
// ----- PART E -----
```

```
averageDifference = generalAverage - specialAverage;
if (averageDifference < 0) averageDifference = -averageDifference;
```

```
double d1 = closestValue - generalAverage;
if (d1 < 0) d1 = -d1;
double d2 = closestValue - specialAverage;
if (d2 < 0) d2 = -d2;
```

```
finalScore = d1 + d2 + averageDifference;
```

```
cout << "Difference Between Averages: " << averageDifference << endl;
cout << "Final Score: " << finalScore << endl;
```

```
deleteAtPosition(list, closestPosition);

cout << "ArrayList After Deletion: ";

displayList(list);
```

```
int roundedSpecial = (int)(specialAverage + 0.5);

insertAtBeginning(list, roundedSpecial);

cout << "Final ArrayList After Insertion: ";

displayList(list);

return 0;

}
```

```
Initial ArrayList: 18 7 45 11 36 31 21 13 29
Minimum Value: 7
Maximum Value: 45
Sum: 211
Median Value: 21
General Average: 23.4444
Special Average: 24.3333
Closest Value: 21
Position of Closest Value: 6
Difference Between Averages: 0.88889
Final Score: 6.66667
ArrayList After Deletion: 18 7 45 11 36 31 13 29
Final ArrayList After Insertion: 24 18 7 45 11 36 31 13 29

-----
Process exited after 0.3876 seconds with return value 0
Press any key to continue . . . ■
```